Research question and mixed research design

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Milestone 1 – The first Assignment

1) **Description and justification of research topic and question**
2) Description and justification for central datasite(s)

*Please submit 1-2 pages to your TA's e-mail by **April 25th**, at the latest, by 11:59 PM.*

This weeks to do list

TUESDAY

- Get into groups.
- Find a topic/question you would like to work on.
- Explore literature for inspiration and knowledge on the topic/question
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THURSDAY

- Continue research on the topic
- Formulate a tentative research question
- Prototype your mixed method design.
  - What data types do you need? (Data sites/source)
  - What types of analysis?
  - What is the reason for mixing data types and analysis types?

"Theory and Design" in Mixed Methods: A Primer for Sociologist.

Frederiksen, Morten, Jonas Toubøl, and Hjalmar Bang Carlsen.

Why Mixed Methods?

- Mixed methods are used to capture the complexity of a phenomenon when single methods fall short.
- The choice of methods should be guided by theory
  "A mixed methods design therefore starts with the theoretical part of the study, where the problem is defined" (p.51)
- Effective mixed methods design begins with a strong theoretical framework that supports the use of multiple data types and analysis methods.

Types of Research Questions

Having a set of theories is not enough for investigation. To design a study, you must develop research questions that operationalize these theories into something that can be empirically illustrated with data.

	Theory testers	Theory developing
Descriptive	1. Is X characterised by Y?	2. What characterises X? What is X associated with?
Explanatory	3. Does change in X cause change in Y?	4. What causes Y? How does X cause change in Y?

Descriptions aim to gain more knowledge about a phenomenon.

- Example: "What are the characteristics of poverty in this community?"

Explanation is about understanding how a phenomenon occurs or changes.

- Example: "How does economic policy influence poverty rates?"

Design Integration

Mixed methods design is about how sub-studies relate to each other.

- **Complementarity:** illuminate different parts of a phenomenon using different data to analyze different aspects
- **Confirmatory:** test if results from different data sources agree - Analyze the same aspect using different data
- **Sequential:** the different parts (or "sub-studies") of your research are done one after the other, not at the same time. The key idea is that the results from the first part influence or inform what happens next.
- **Simultaneous:** sub-studies are conducted at the same time, with results integrated only after completion

Design Weight

Not all data sources play equal roles in mixed methods research.

- **Primary Study**: carries more analytical weight in final conclusions
- **Secondary Study**: provides supporting evidence or context

CHAPTER 3

Theory and design

This chapter will focus on answering three key questions:

- Why should you choose a mixed methods study?
- What is integration in mixed methods design?
- Which mixed methods design should you choose?

The chapter begins with some general methodological considerations about how we in the social sciences approach our research subjects. Here we further discuss the relationship between theory, evidence and data, which is fundamental to all research, but which becomes particularly central to mixed methods studies. The reason is that in a mixed methods design, it must be very clear how different types of data and analytical methods are connected to the theoretical starting point. This is what we called theoretical integration in chapter 2: that we have a clear theoretical understanding of the phenomenon we are investigating, and that the theory justifies the necessity of collecting different types of data and evidence to shed light on the theoretical ideas about the phenomenon.

It's also the answer to the first question: you should choose to conduct a mixed methods study in order to capture all the aspects of a phenomenon that theory emphasises as important, and not just focus on the parts that one type of data is good at capturing.

A key point in this chapter is therefore that a problem can and should often be operationalised into several research questions to take into account the relevant dimensions of the theoretical basis of the research.

Intended learning outcomes

Knowledge

- Show familiarity with the basic techniques, use scenarios, and validity criteria of computer-assisted qualitative methods, i.e. digital ethnography, content analysis, and exploratory network analysis.
- Account for the procedures, potentials, and pitfalls of combining qualitative and quantitative data sources, including in integrated quali-quantitative ways.
- Account for the relationship between digital methods' emphasis on the media contexts of digital data and the broader questions, claims and biases of social data science.

Skills

- Identify the procedures of qualitative content analysis for designing appropriate semantic categories, including for use in subsequent machine learning with quantitative text (and/or visual) data.
- Extract, and communicate patterns of networks, ideas, and behaviour characteristic of specific social settings and public issues, using the appropriate qualitative method(s).
- Combine qualitative data with a quantitative data source, thereby integrating heterogeneous digital data formats into comprehensive social analyses.

Competencies

- Evaluate and analyse a social data problem from both qualitative and quantitative perspectives, including determining when to deploy which method designs.
- Design and implement small-scale digital ethnography campaigns, along with exploratory network analysis and content analysis, to obtain insights into social networks, ideas, and behaviour at individual and organizational levels.
- Combine qualitative and quantitative sources of data, as well as forms of narration and visualization, into persuasive quali-quantitative reports on social data problems for a range of organizational use scenarios.